Smart Waste Classifier

AI-powered Garbage Image Classification using ResNet18 and Vision Transformer

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Introduction & Problem Statement

Waste segregation is a critical global challenge, directly impacting environmental sustainability and resource management. Inefficient sorting leads to landfill overflow and pollution.

AI offers a powerful solution by automating and enhancing waste classification, making recycling processes more efficient.

Objective

To develop a robust classification model capable of automatically identifying various types of garbage, including paper, plastic, metal, and organic waste, from image data.

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Data Exploration

Our project utilises a comprehensive image dataset featuring diverse waste categories, crucial for training effective classification models. The dataset comprises over 2,500 images across 6 distinct waste classes.






Each class is represented by a variety of images, ensuring the model learns to identify waste types under different conditions.

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Data Preprocessing

Effective data preprocessing is vital for optimal model performance. Our pipeline involves several key steps to prepare the image dataset for training.

1

Image Resizing

All images are uniformly resized to 256x256 pixels.

2

Normalization

Pixel values are normalized to a standard range (0-1).

3

Data Augmentation

Techniques like random rotations, flips, and colour jittering are applied to enhance model robustness.

4

Data Loaders

Efficient data loaders are used for batch processing and parallel loading.

5

Train/Val/Test Split

Dataset split into 70% training, 15% validation, and 15% testing.

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Model Development & Training

We leveraged two distinct deep learning architectures, ResNet18 and Vision Transformer (ViT-Tiny), to evaluate their performance in waste classification.

ResNet18 (CNN)

A convolutional neural network (CNN) known for its residual connections, mitigating vanishing gradients in deep networks.

- Epochs: 20
- Learning Rate: 0.001
- Optimizer: Adam
- Loss Function: Cross-Entropy

Vision Transformer (ViT-Tiny)

A transformer-based model that processes images as sequences of patches, capturing global dependencies.

- Epochs: 20
- Learning Rate: 0.001
- Optimizer: Adam
- Loss Function: Cross-Entropy

Both models were fine-tuned using the timm library in PyTorch, ensuring optimised training and reliable performance metrics.

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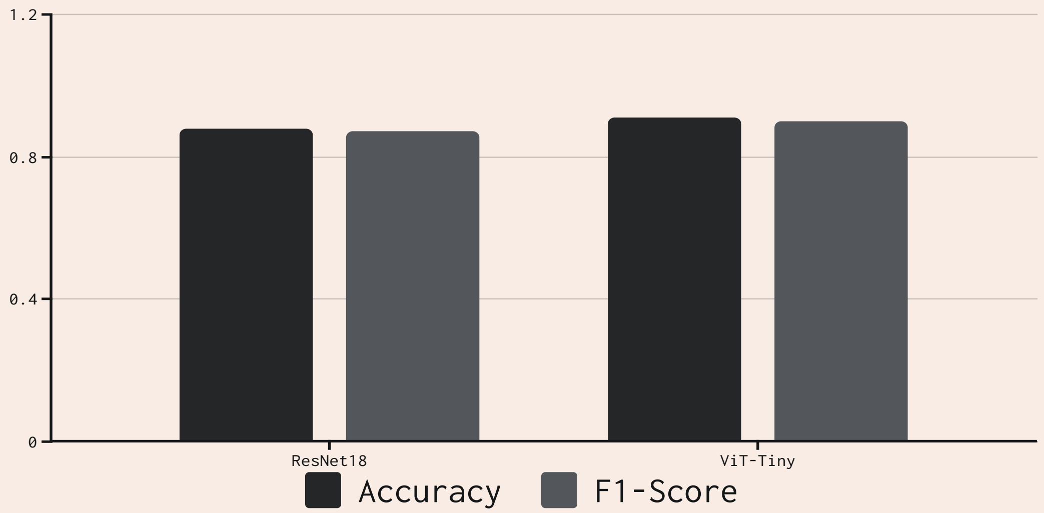
Evaluation and Comparisons

Rigorous evaluation using key metrics provided insights into model performance and architectural strengths.

Key Metrics

- Accuracy: Overall correctness
- Precision: Correct positive predictions
- Recall: Identified actual positives
- F1-Score: Harmonic mean of precision and recall

Confusion Matrices: Visual representations for detailed per-class performance.



ViT-Tiny generally outperformed ResNet18, demonstrating the power of transformer architectures for image classification.

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Classification on Unknown Data

Our models are designed to seamlessly classify new, unseen images, providing instant waste identification.

Prediction Process

Unknown images are input into the trained models (ResNet18 or ViT). Each model processes the image and outputs a predicted class label, along with a confidence score indicating the certainty of the prediction.

Example: An image of a plastic bottle is predicted as 'Plastic' with a confidence of 98%.

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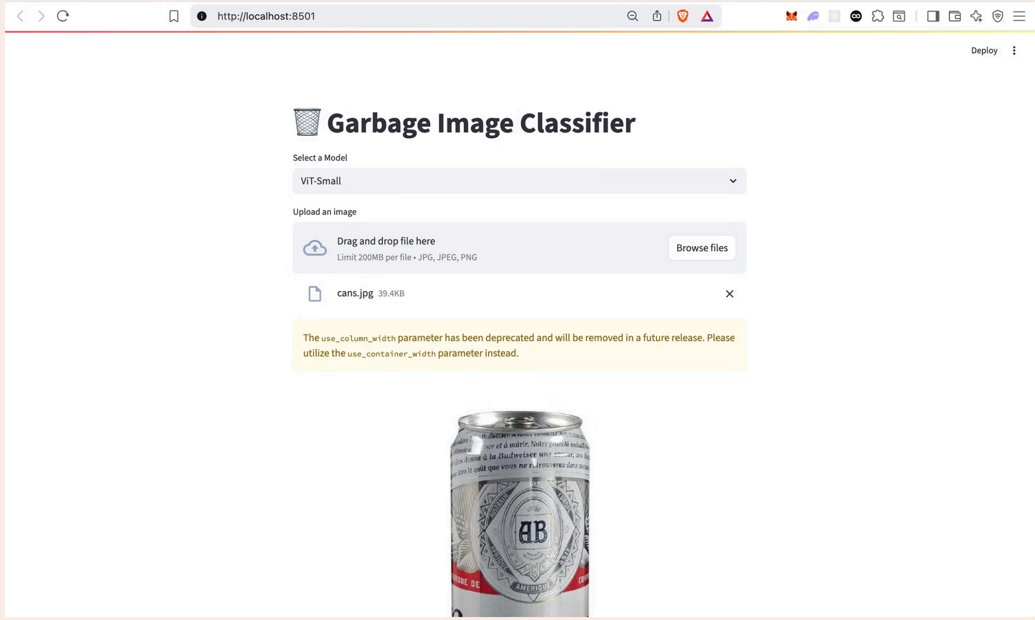
Streamlit App Deployment

To make our classification models accessible and user-friendly, we developed an interactive web application using Streamlit.



App Features

- **Image Upload:** Users can upload waste images directly.
- **Model Selection:** Choose between ResNet18 and ViT for classification.
- **Real-time Prediction:** Displays predicted class and confidence metrics.

Technologies Used

- Python
- Streamlit
- PyTorch
- timm (PyTorch Image Models)

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Lessons Learned & Reflections

Ashwin Manickam Nagappan

Improved understanding of CNNs and model optimisation.

Athika Fatima

Streamlit deployment and ViT architecture integration using timm.

Osemudiamen Iyamah

Evaluation metrics and confusion matrix interpretation.

Mohammed Wali Uddin Qureshi

Data augmentation and handling class imbalance.

Agha Mohammed Hussain

Debugging model loading errors and ViT compatibility.

Claude Sylvain Baumono Pugueu

Real-world application of AI and collaboration in ML pipeline.

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